

Sublevel Stoping

30/12/2019

Principle I

- *Sublevel stoping is also known as blast hole or long hole stoping.
- *Sublevel stoping is an open stoping method.
- *It is high production, bulk mining method.
- *Sublevel stoping accounts about 3% of world non coal production.

Ore Body I

- *Applicable to large, steeply dipping regular ore bodies.
- *Ore, HW & FW should be strong to fairly strong and competent, self supporting.
- *Ore strength should be 55 Mpa or more.
- *Ore body is required to be studied well in advance to avoid surprises at the time of mining.
- *Variation of rock strength should be compensated at design stage.

Ore Body II

- *Dip of ore body, foot wall must be such that it exceeds angle of repose of broken ore.
- *This shall permit gravity flow of blasted ore through draw point and chute.
- *Ore body should be more than 6m wide. 3

Ore Body III

- *Less than 6 m width long hole drilling and blasting is not performed because of higher cost and larger dilution.
- *In wide, relatively weaker ore bodies, transverse stoping is applied.
- *Support pillars (rib pillar) are left in ore body, which are removed after adjacent stope being filled up cemented fill.
- *Ore body should be even and fairly well defined.
- *Stope boundaries must be regular.
- *Irregular ore boundaries or large waste inclusion caused dilution of ore.

Ore Body IV

- *Ore body should have smooth foot wall contact.
- *Smooth FW contact allows flow of blasted ore to draw points and chutes.
- *Rock (ore, FW & HW) must be structurally competent and self supporting.
- *Larger opening is left unfilled for extended period of time.
- *Repeated shocks from large blast requires ore of high compressive strength and minimal structural weakness, such as joints, fault and bedding plane.

Ore Body V

- *Failure of HW / stopes back causes excessive dilutions, loss of sublevels and large chunks blocking draw points.
- *Small, localized ground failure results in ground movement and displacement of blast holes.
- *Ground failure makes loading of blast hole difficult and in some cases requires extensive drilling in stoping block.
- *Sublevel stoping can be used up to 1km depth.
- *At great depth, stope pillars are crushed and tend to fail, causing sub level stoping difficult.

Stope Development I

- *Mine development starts from a shaft sunk in foot wall to avoid any subsequent caving effect from stopes.
- *Ore body is developed vertically by driving crosscut and haulage levels every 45 to 120m.
- *Access raises driven in ore body are used to further subdivide ore body into blocks for stoping.
- *Collection system is constructed by undercutting stop blocks.
- *Sublevels are driven through proposed block every 10 to 55m.

Stope Development II

- *Ramp from lower level is excavated up ward at 9 degree to reach upper sublevel.
- *Ramp provide better access to sublevels from main level.
- *Ramp improves machine & stope utilization.
- *Ramp helps to provide better supervision by allowing supervisor to move with vehicle.

Stope Development III

- *More than one sublevel may be used at each levels, depending on width of ore body.
- *Slotting raises are excavated from one haulage level to upper haulage level.
- *Blasting long holes, slotting raises are converted into expansion slot to height and width of proposed stope.
- *Shrinkage stopes has also been used to from starting slot.

Stope Development IV

- *Slotting is very expansive part of stoping operation.
- *25 to 30% cost of mining cost consist of development & slotting.
- *Stoping is carried out by blasting vertical slices in to an expansion slots.
- *Stopes are contained by crown pillar, which protects level above, rib pillar and sill pillars, through which ore collection system is cut.
- *Rib pillars are some time replaced by filled stopes.
- *Pillars are removed through large scale blasting, when stope is exhausted.

Stope Development V

- *Up to 100% of ore can be recovered under ideal condition.
- *Stopes are to be filled, when caving is unacceptable and minimize underground stress.
- *Filling of stopes also facilitate pillar removal.

Stope Extraction System I

- *Base of stope is slotted (above sill pillar) horizontally prior to production blast.
- *Base of stope (sill pillar) is excavated to make 45 degree collection trough.
- *Some time, no slot is excavated above sill pillar, Stope blasting exposes collection system as stope face retreats.
- *Excavation of trough or cone is typically carried out by drilling from draw points and mining up wards using (55 to 94 mm dia long hole)
- *Distance between base of stope and haulage level varies from 9 to 21 m (sill pillar thickness).

Ore Handling at Draw Point I

- *Most modern system uses LHD (tire Mounted) loading directly out of draw point. LHD then travel to ore pass. Remote controlled LHD has proved most successful for loading out draw points under exposed condition, particularly near end of stope life.
- *LHD in turn load ore on to LPDT for hauling to ore pass.
- *Longer hauling of ore is done by track mounted cars.

Production Drilling I

- *It is required to established drill ability of ore, required fragmentation, size, hole size, hole length, orientation, spacing of drill holes and drilling accuracy for drill hole pattern design.
- *Drill machines is based on these information.
- *Production form sub level stope is by blasting long holes.
- *Hydraulic Drill machines mounted on truck has become common for faster drilling & high level of automation.

Production Drilling IA

- *Mostly DTH drill machines are deployed for drilling long holes.
- *Machines are mounted on column and arm.
- *

Production Drilling II

- *High pressure pneumatic DTH hammers or rotary percussive drilling system can drill up to 229mm dia (9inch) hole up to 120m length with less than 2% deviation.
- *Drilling accuracy has economic implication in mining operation.
- *Many drill machines has 360 degree ring drilling holes, includes up and down holes.
- *Best drilling and blasting preference is obtained, when vertical holes are drilled from drill sill developed in sublevel exposing HW and FW of stope.

Production Drilling III

- *Ring or parallel hole are drilled for sub levels which are 18 to 120m apart.
- *Parallel holes are drilled for steeply dipping deposit.
- *Ring pattern drill is followed in massive ore bodies.
- *With introduction of up hole and down hole drilling rig, drill sill no longer to be developed for HW & FW.
- *Drilling pattern and hole sizes are frequently established by pre setting powder factor to size if ore block to be mined.
- *Typical hole spacing varies from 2m to 6m and burden varies 1m to 2.5m.

Production Blasting I

- *Selection of explosive is done based on requirement in fragmentation, hole size, spacing and burden, holes conditions, water inflow, allowable blast size, hardness etc.
- *ANFO is least expansive explosive.
- *ANFO can be freely poured in down holes or pneumatically loaded in ring holes.
- *Up holes up to 150mm diameter can be charged by ANFO by specially constructed collar plug and cascading technique.
- *ANFO can not be used, where water flow is a problem.

Production Blasting II

- *ANFO can not be allowed to be in watery hole for a period more than few hours.
- *Emulsion explosive can be used for watery hole.
- *Emulsion explosive is water proof, high density and typically high velocity explosive.
- *Emulation explosive can be loaded in down holes or up holes.
- *Column loaded holes are simply loaded from toe to collar with explosive.
- *Deck loaded holes are loaded with alternating decks of explosive and an inert stemming material.

Production Blasting III

- *Decision to load a hole with either a column charge or deck charge is based on consideration such as allowable powder factors, hardness of ore, up holes and down holes conditions, fragmentation requirement, blast vibration.
- *Special charges is used to blast large diameter down-holes and charge height equal to six bore hole diameter maintained with in hole (deck charging).
- *Blasting is sequential, so that one ring of holes is fired at a time,
- *Rings are blasted in succession in to opening created by blasting of previous ring.

Production Blasting IV

- *Timing sequence of blast is critical to success of blast and resultant fragmentation.
- *Proper blast timing is critical in minimizing the effect of excessive blast vibration and concussion effect underground.
- *Secondary blasting is necessary if primary blasting has failed to sufficiently fragment ore to that can be dealt with ore removal and hoisting equipment.
- *Plaster shooting (Shaped charge) involves placing explosive on top of bolder and boulder is shattered by detonating explosive.

Production Blasting V

- *Secondary blasting is an expansive, time consuming exercise.
- *Close engineering control on primary drilling and blasting will minimize frequency of secondary blasting.
- *Blasting cost is 10% of total mining cost and drilling costs 30% of total mining cost.

Back Filling I

- *Large opening created by sublevel stoping typically requires back filling.
- *Back fill includes rock and sand fill, cemented rock fill, cemented hydraulic tailing fill, high density tailing or alluvial fill.
- *Back fill allows for future recovery of support pillar.
- *Recovery of pillars leads to 90% recovery of ore.
- *Back filling minimizes occurrence of subsidence & allows for re-distribution of stress created by mining cycle.

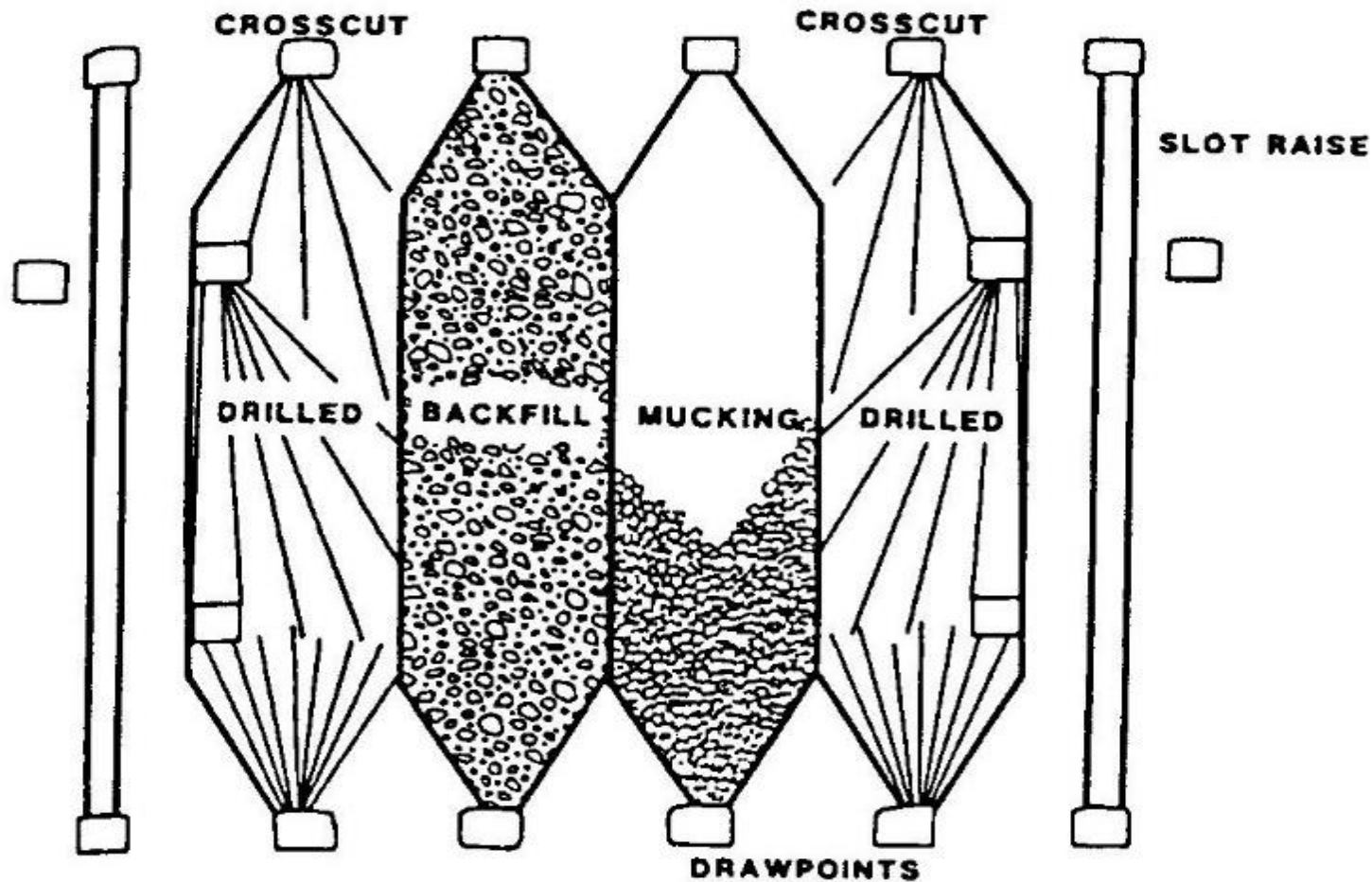
Back Filling II

- *Back filling minimizes frequency of rock bursting.
- *Back filling has also been used successfully to eliminate rib pillars between stopes.
- *Back fill should have sufficient cement to form a self supporting unit.
- *Cemented back fill is not economical when pillar recovery is not possible.
- *In case ore pillars are not to be recovered, ordinary back fill (non cemented) is used for controlling subsidence.

Back Filling III

- *Back filling, by mill tailing also reduces volume & surface land requirement for tailing pond.
- *Underground disposal of tailing also reduces pollution load on surface.

Sublevel Stopping



**NO. 2 MINE LONGITUDINAL SECTION
THROUGH MINING BLOCKS**

Fig. 18.4.4. Cemented backfill used as alternative to pillars
(Belford, 1981).

Back Fill Sublevel Stopping

Stopes are cemented back filled for pillar recovery.

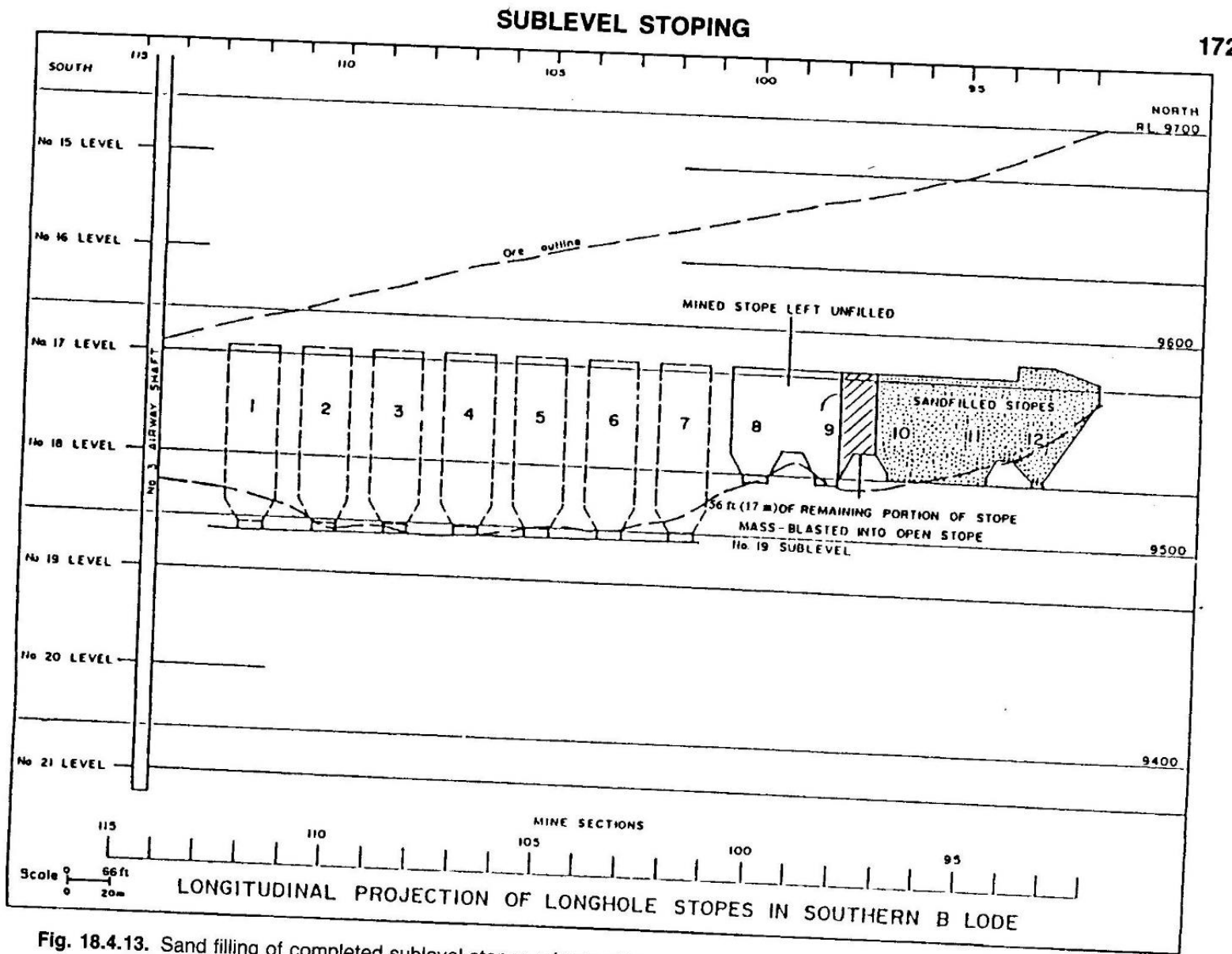


Fig. 18.4.13. Sand filling of completed sublevel stopes prior to pillar recovery (Smaniotto, Solomons, and Tillmann, 1981).

Support I

- *Ground control is minimized with sub level stoping, because rock must be inherently strong.
- *Different type of rock bolts and cable bolts are used for local support.
- *Some time regional stress have been shown to effect stope design and dimensions.
- *Regional stress are to be managed by local support.
- *Large open stope can leads to concentrate high horizontal stress and cause severe deterioration in development opening in close proximity of stopes
- *Cable bolting has proved very successful for full stop wall support using cable up to 45m longh.

Support II

- *Cables are normally installed from blasting sub level during blast hole development.
- *Resin anchor bolts are most successful, but cement grout bolts are also used.
- *Variety of cables are used for cable bolting from old hoist rope and low relaxation strand rope / cable.

Support Sublevel Stopping

**Stopes
drives and
other
adjoining
areas are
supported
by Rock
Bolts.**

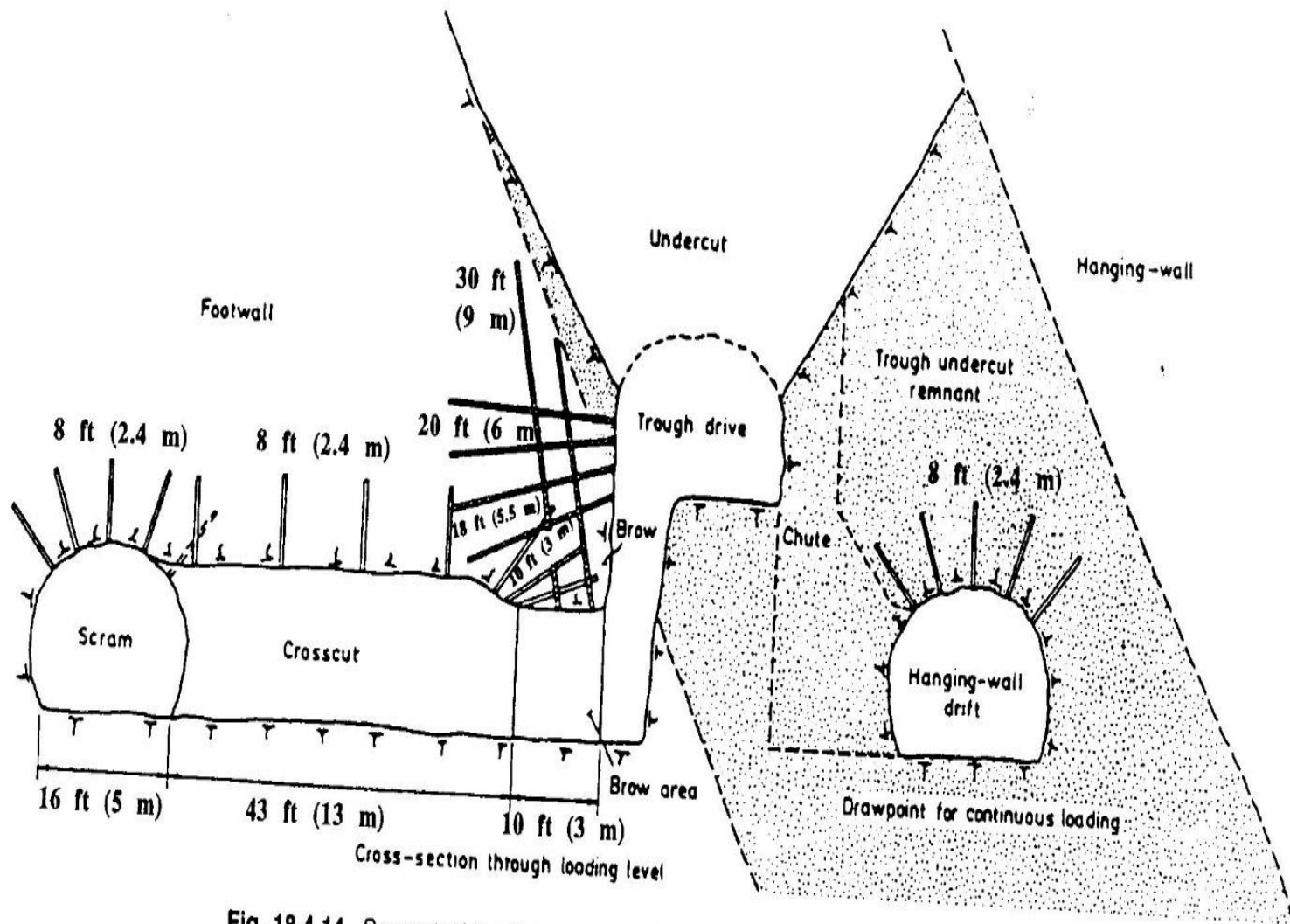


Fig. 18.4.14. Support of loading crosscut showing brow reinforcement (Maw, 1986).

Support Sublevel Stopping

HW of stopes are supported by cable bolts in holes drilled from blasting sub level

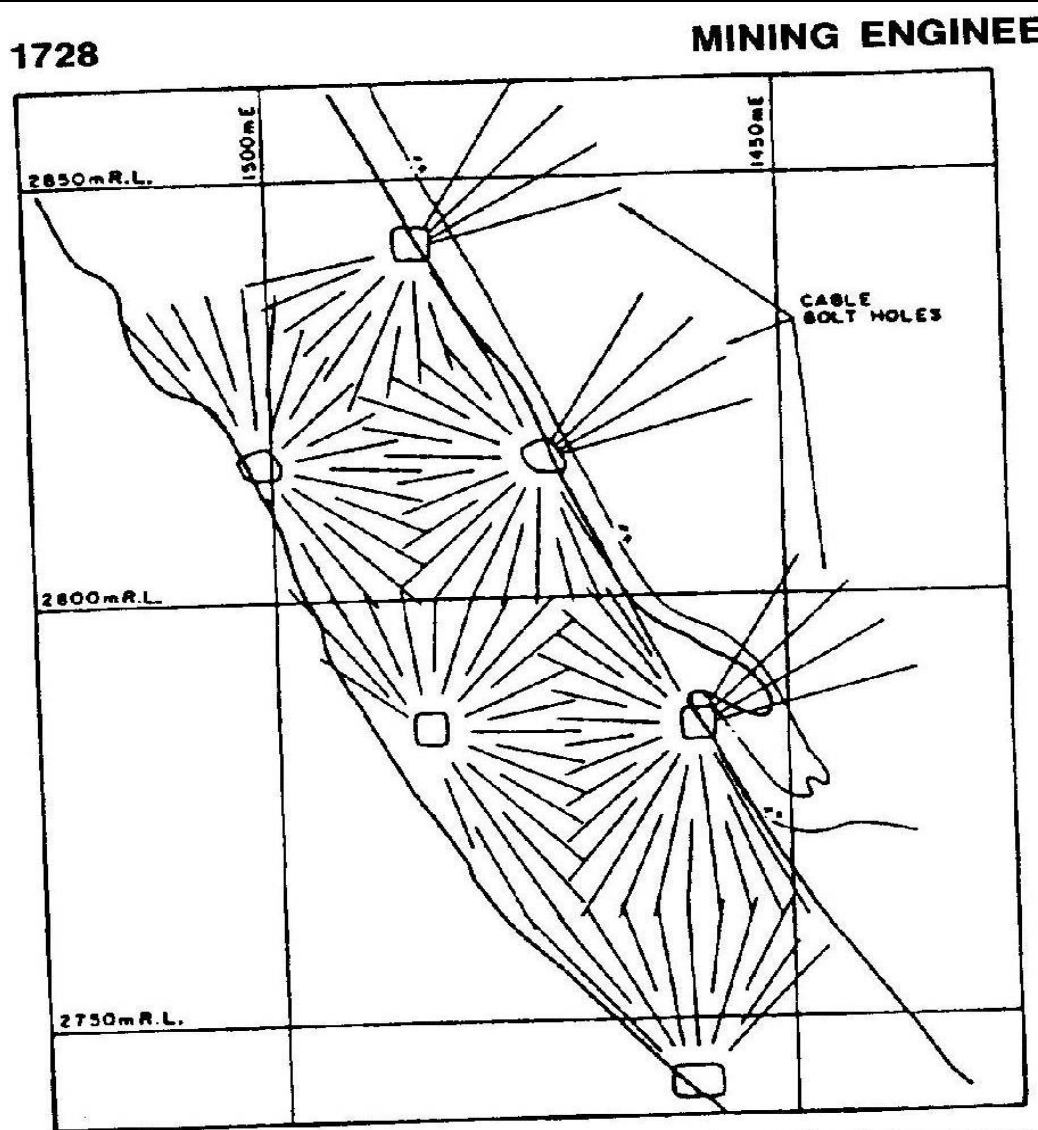


Fig. 18.4.15. Cable bolting holes for hangingwall reinforcement drilled from stope sublevels (Goddard, 1981).

Safety (Advantages) I

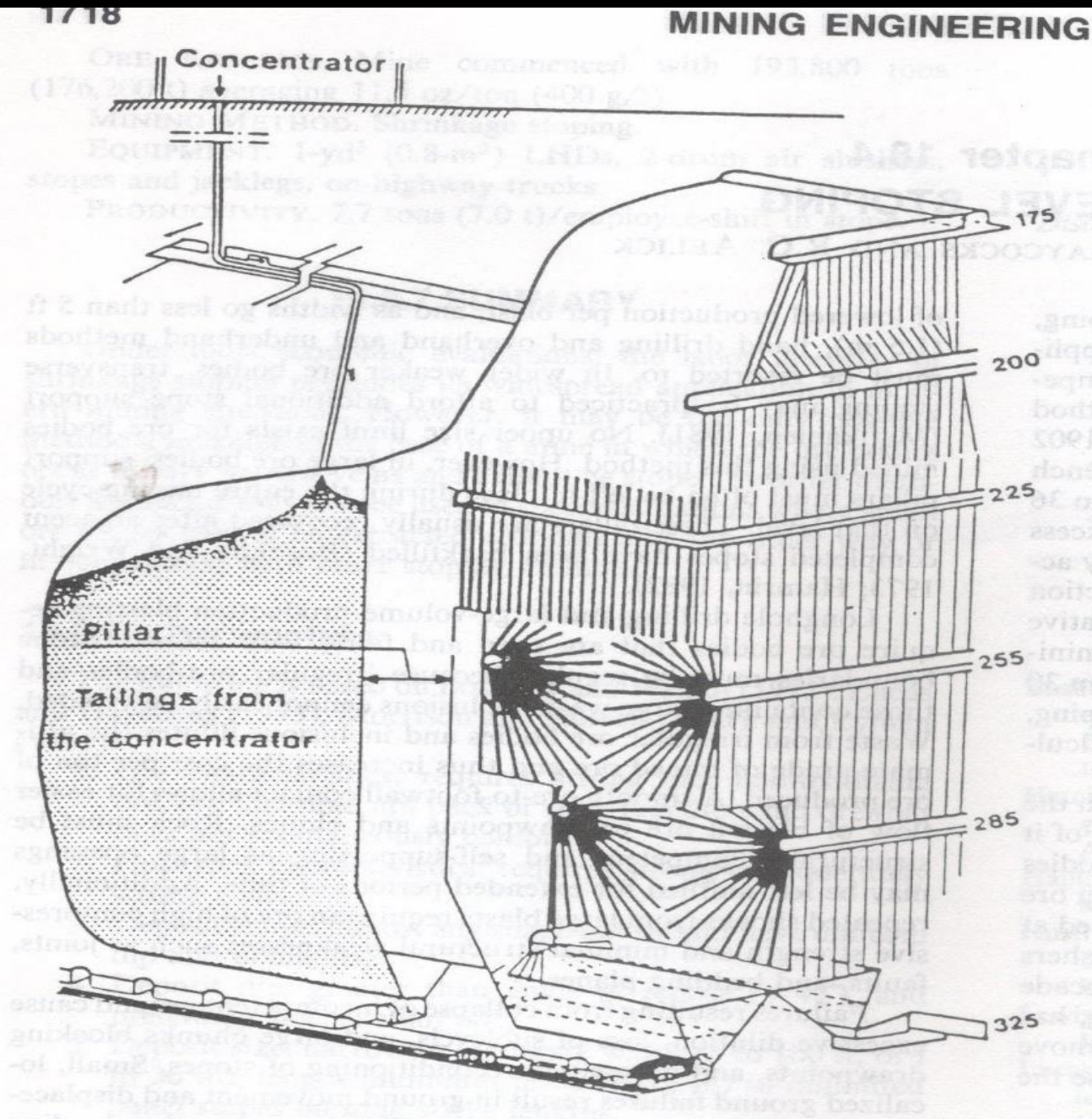
- *Sublevel stoping is very safe mining method by virtue of design.
- *Miner work only under conditioned ground, that has been secured by means of rock bolts, cable bolts and artificial support.
- *Miners are not required to be in broken ground.
- *Mining system is retreating from unsupported to supported area.
- *High level of mechanization helps in reducing exposure of workmen.
- *Remote controlled LHD (cordless control) prevents workers to go to unsupported stope portion.

Safety (Advantages) II

- *Drill Machines can be operated remotely, reducing exposure of driller.
- *Blasting in sub level stoping is done once in a while and blasting is done from safe refuse station of from surface.
- *Workers only enter in stope after all blasting fume are exhausted by means of fan.
- *Long cross section of drift, raise, decline allows large ventilation air flow in mine.

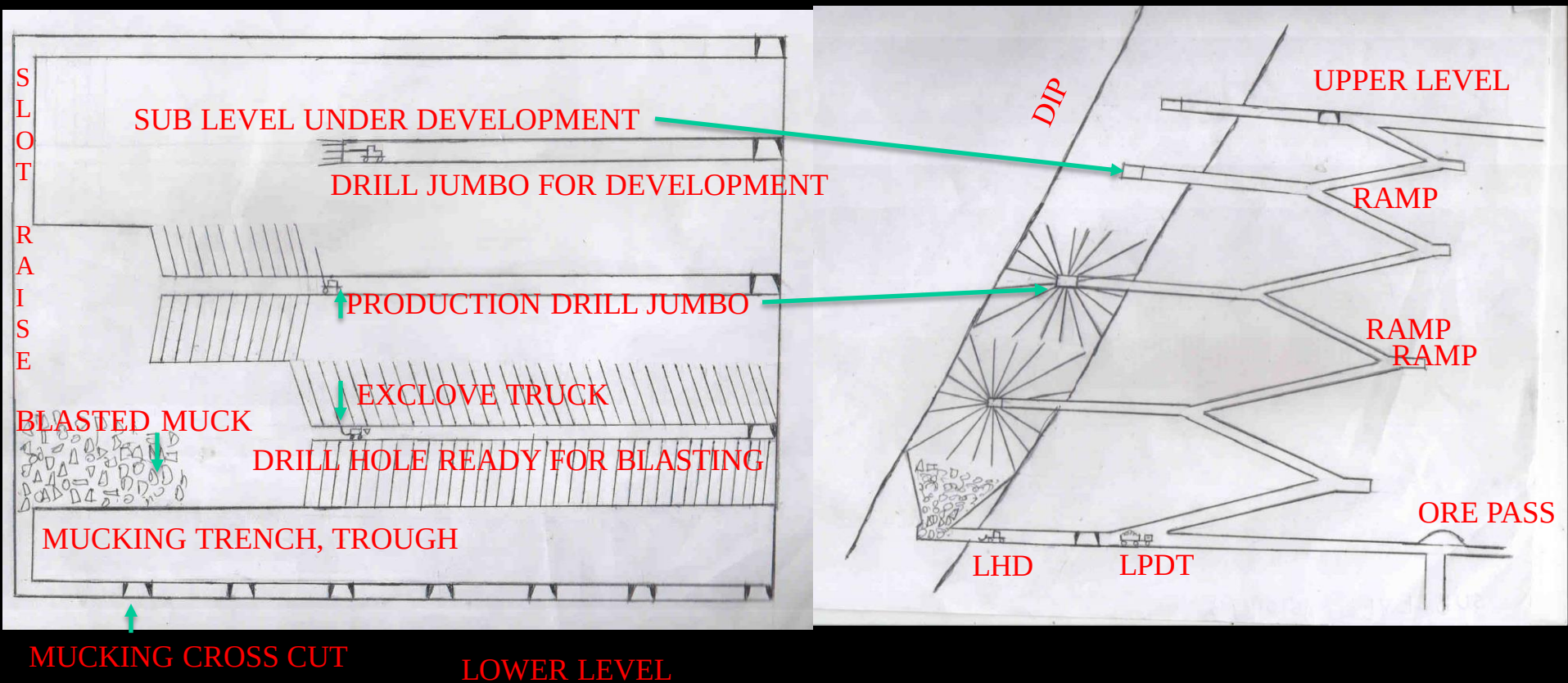
Sublevel Stoping

**Production
Blasting
With
filling**



Mechanised Sublevel Stopping * With LHD, in Trench (Trough) Mucking

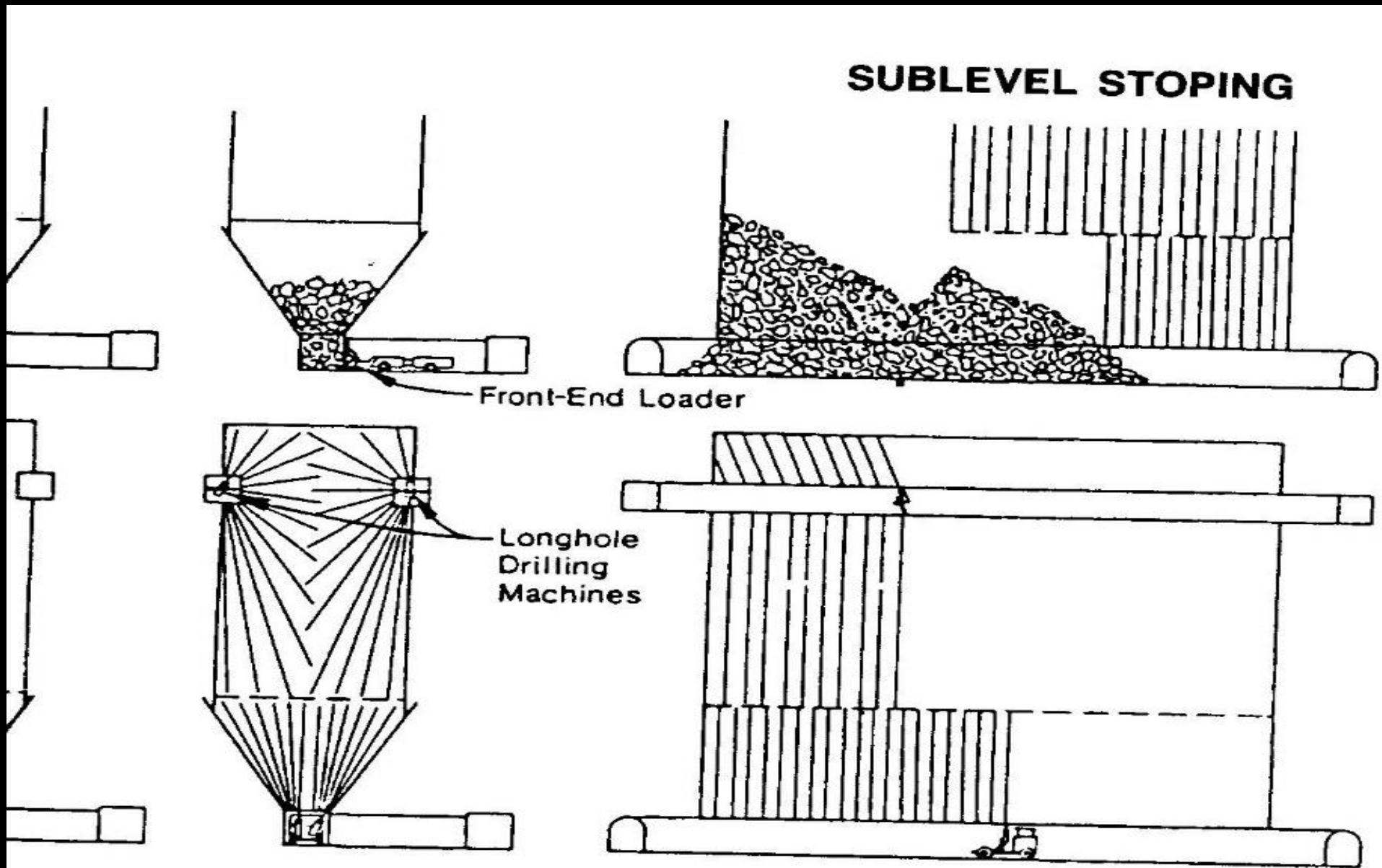
UPPER LEVEL



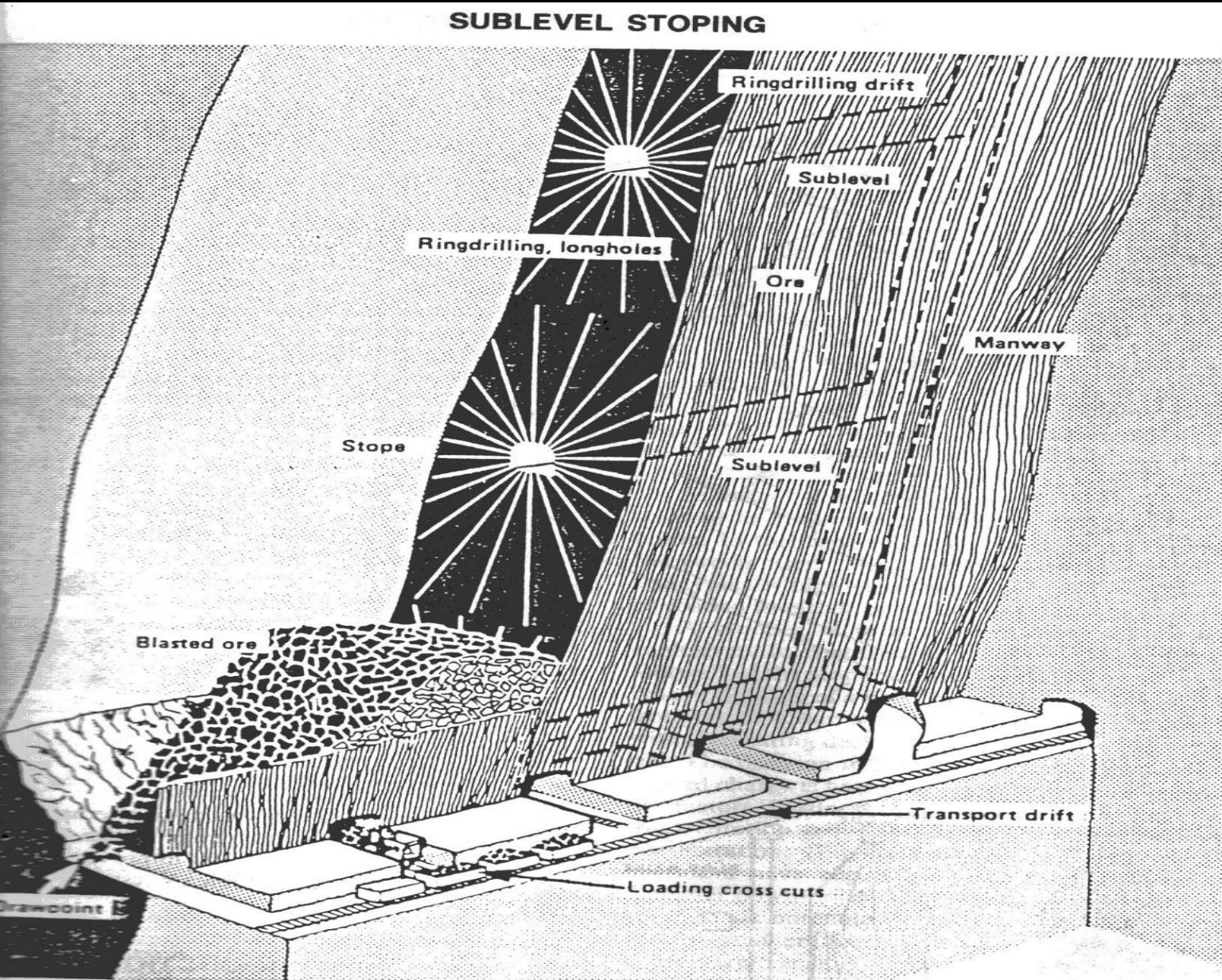
MUCKING CROSS CUT

LOWER LEVEL

Ore Collection by LHD from Draw Point



Sublevel Stoping

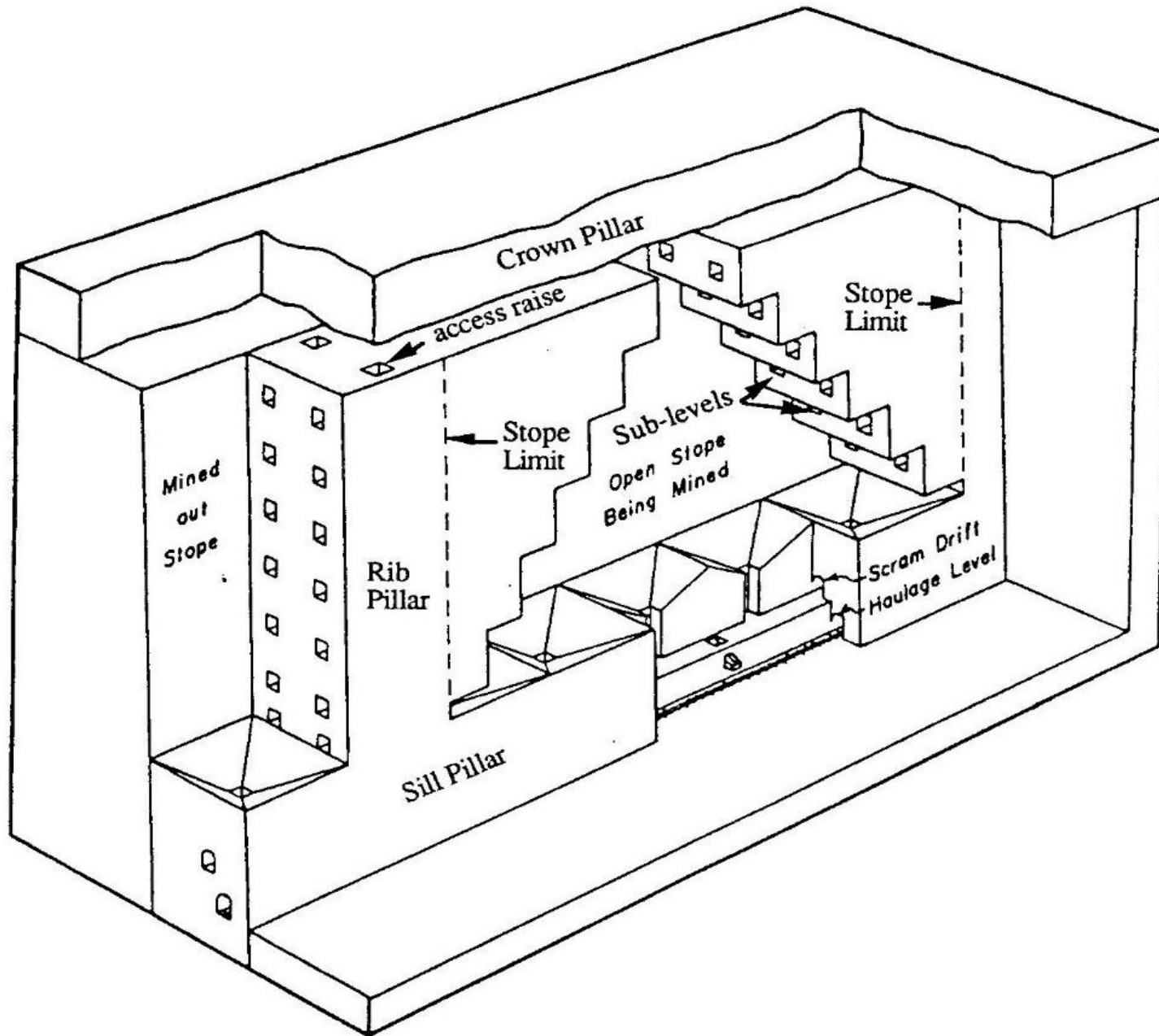


**Production
Blasting
With
Ring hole
Pattern and
loading
system**

L4.2. Production blasting in a sublevel stope showing ring drilling patterns and loading system (Hamrin, 1982).

Sublevel Stopping

**Pillar
system
Sublevel
Over hand
Stope.**



Draw Point Excavation

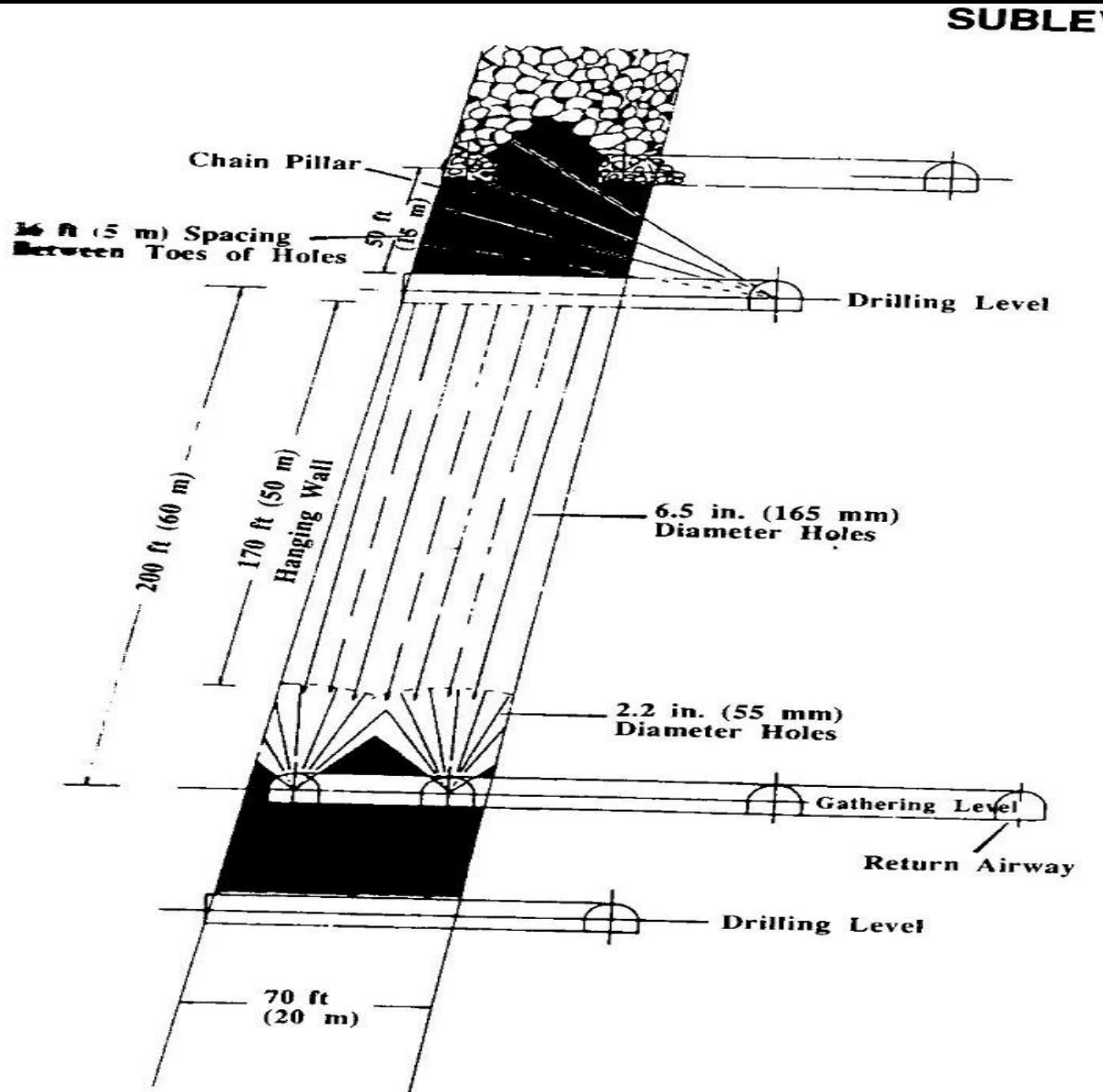


Fig. 18.4.5. Use of small-diameter holes for cutting collection cones (Mabson and Russell, 1981).

Ore Handling at Draw Point I

Non-mechanized

*Ore can be directed from chine through finger raises to grizzly level and then directly to car. Secondary blasting is confined to grizzly it self. This system is usually selected for very large stopes where cost of development is offset by its longevity.

Ore Handling at Draw Point II

Non-mechanized

- *Ore can be drawn directly into finger raises or chutes in case when it breaks very fine and then loaded to mine car.
- *Single box hole and chute feeding into a slusher drift may be used. Up to ten draw points feed in to slusher drift that will pull ore to a grizzly or directly in to car through chinaman.
- *Loading from draw point by track mounted loader to rail haulage system.
- *Ore is directly loaded from draw points in to pan feeder.

Ore Collection by Ore Pass

Ore
directly
loaded
trough
Grizzly on
to car

1722

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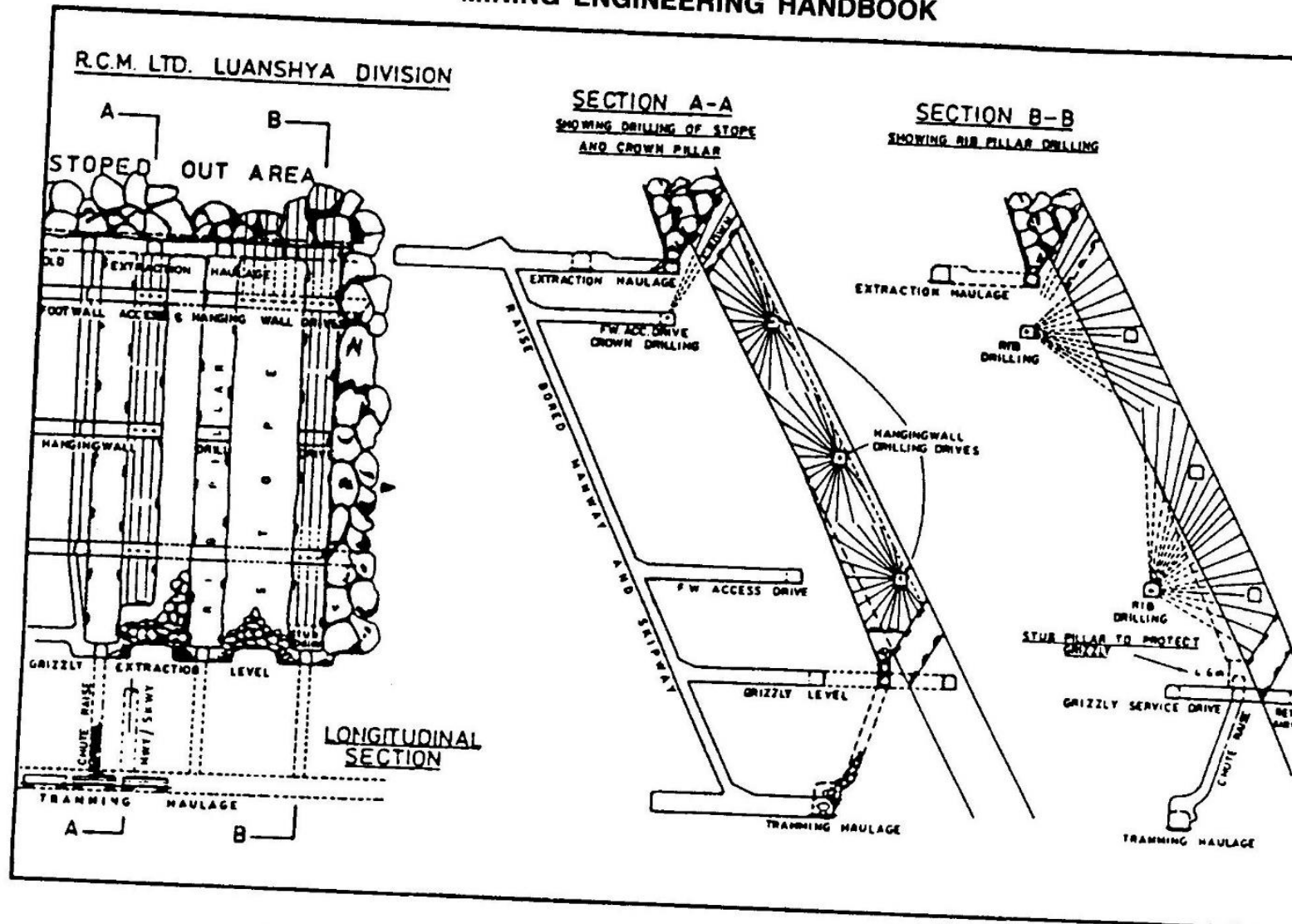
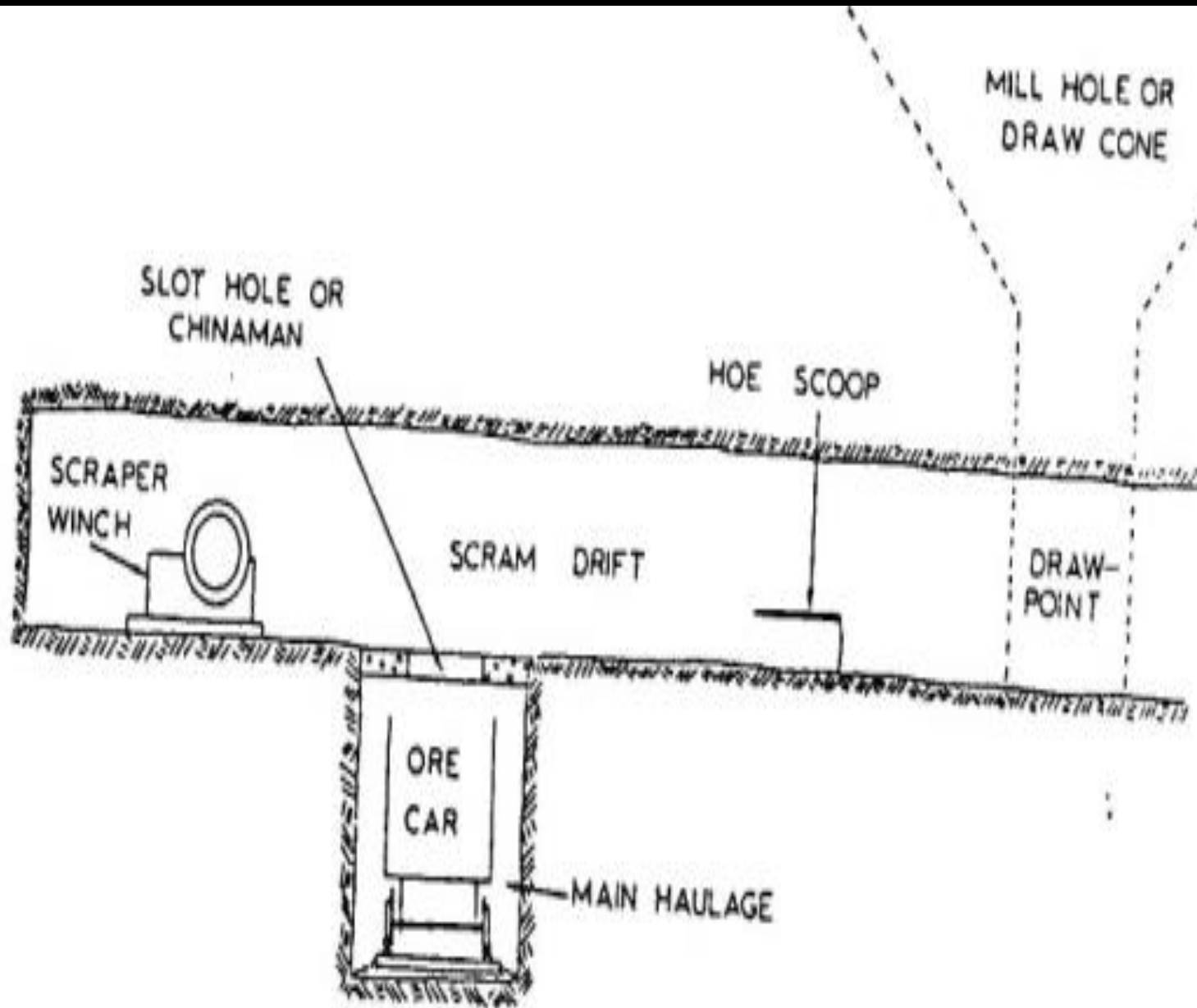


Fig. 18.4.6. Grizzly and orepass collection system (Mabson and Russell, 1981).

Ore Collection Directly on Car

**Using
Scraper
ore Directly
loaded on
car**



Ore Collection by Line Loader

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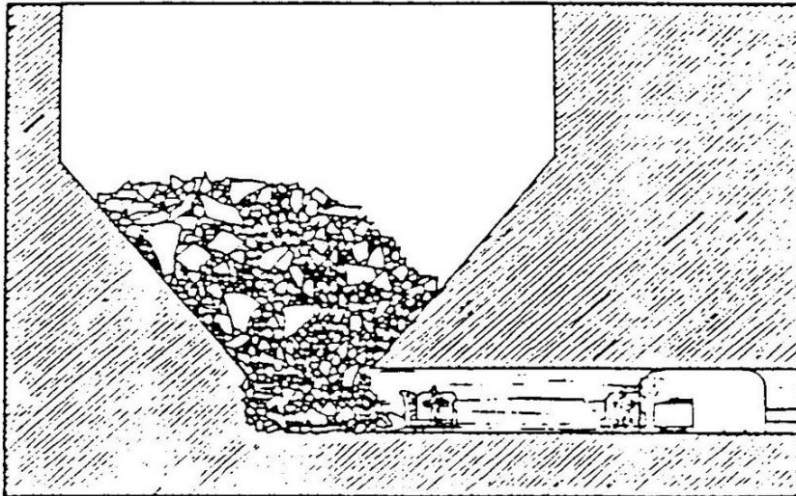
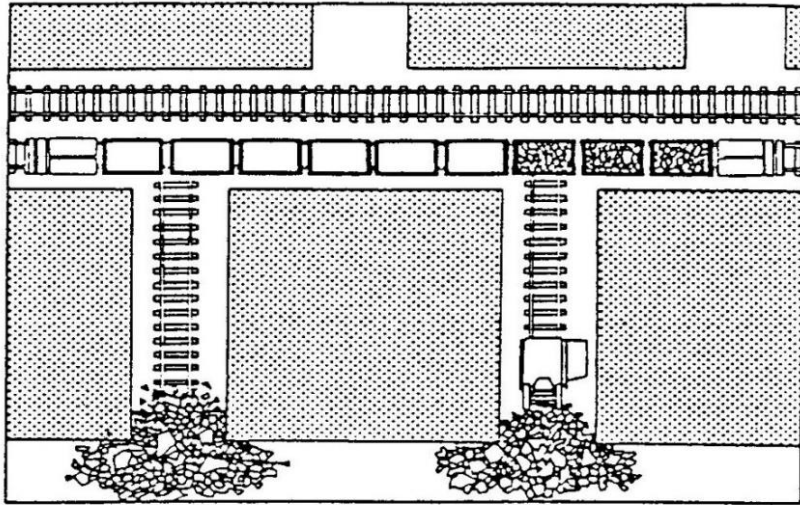


Fig. 18.4.9. Track-mounted loading system for sublevel stope (Bjorkstedt, 1981).

Under Hand Sublevel Stoping I

- *Stoping starts by blasting topmost most sublevel face, then moves progressively down wards.
- *Individual sublevel is blasted by upper hand drilling.
- *Face blasting is carried out vertically.
- *Mucking is done from bottom of stope draw point at sill pillar.
- *Under hand sublevel is practiced for improving stability of stope.

Under Hand Sublevel Stoping II

*Underhand sublevel is safe for weak ore body as man are working at well supported sill pillar and level below.

*Sill pillar supported by ore body below, remains in better shape under pressure.

*Under hand sublevel method has mucking problem.

*Uranium, Fluorine-Fluorite Mine, Colorado, USA

Under Hand Sublevel Stopping

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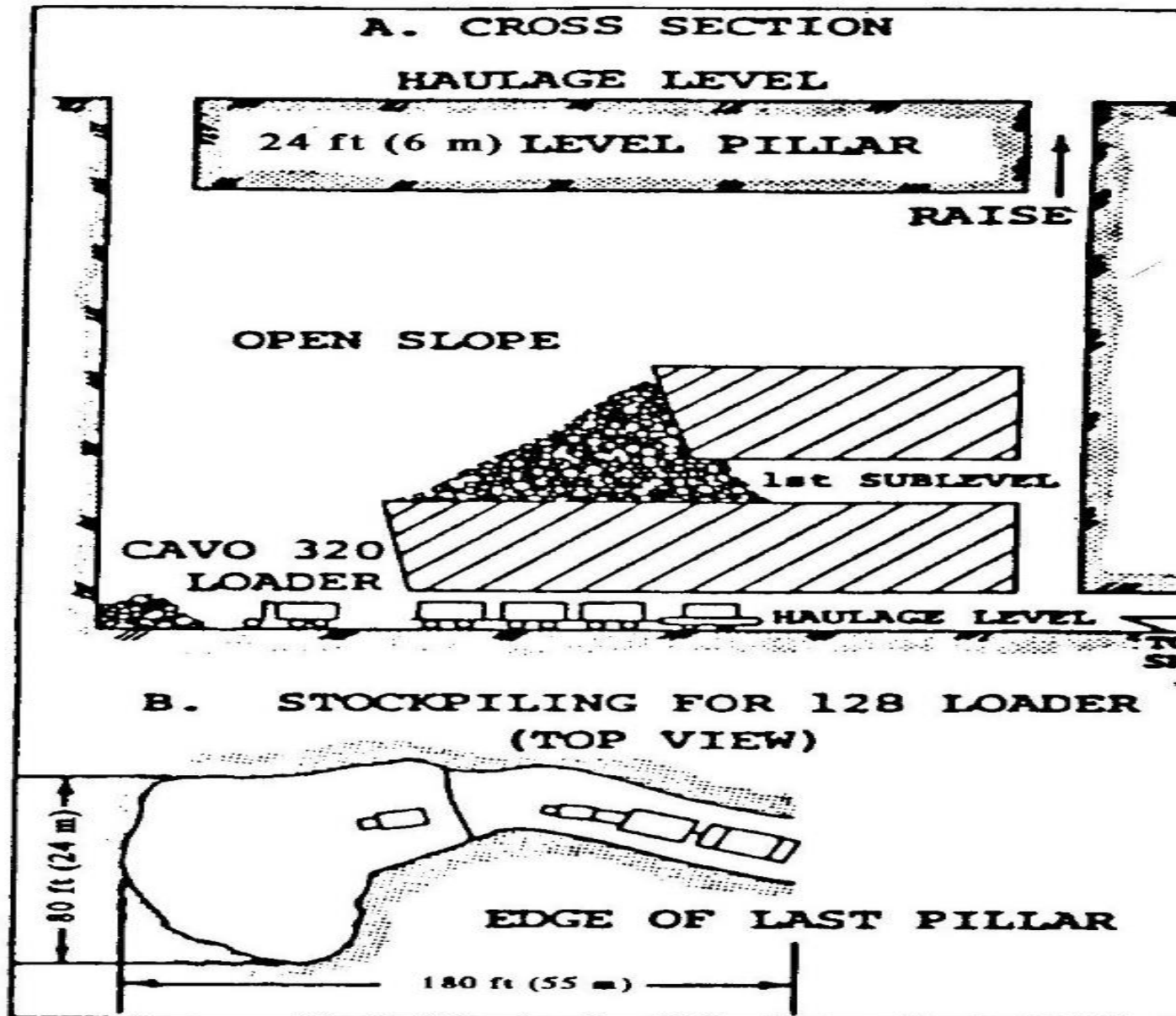


Fig. 18.4.10. Sublevel faces carried underhand (Lawrence,

Advantages I

- *Sublevel stoping is very amenable to mechanization, stoping efficiency is very high.
- *Sublevel stoping is high production, low cost mechanized method
- *Sublevel stoping is safe and easy to ventilate.
- *Ore recovery is high, 90% with pillar recovery.
- *Blasting is done periodically and highly skilled blasting crew.
- *Ore production starts immediately after blasting.

Advantages II

*Sublevel stoping is primarily under ground method, when surface mining of the deposit is no longer economical & surface subsidence is not allowed .

Disadvantages I

- *Sublevel stoping is capital intensive (large development cost)
- *Sublevel stoping is not selective, most of ore has to be mined.
- *Sublevel stope is very inefficient at low dip (dilution goes up significantly).
- *Secondary may require evacuation of stopes for not exposing worker to blasting fume.
- *Development cost is 30% of total mining cost.
- *Sublevel stoping is not suitable at depth more than 1km.

References I

*SME Mining Engineering Hand Book 2nd Edition,
page 1717 to 1731